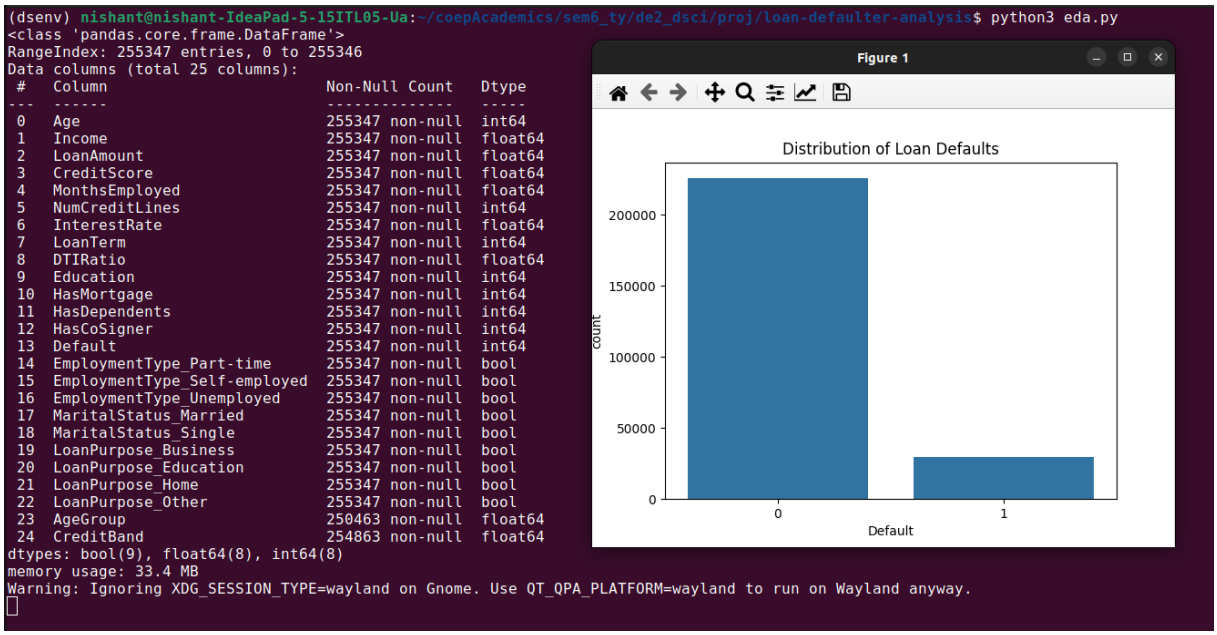
Exploratory Data Analysis

Dataset Overview

The dataset contains approximately **255,000 loan records** with multiple demographic and financial attributes such as income, loan amount, interest rate, debt-to-income ratio, and employment status. The target variable **Default** indicates whether a borrower defaulted on the loan (1) or repaid successfully (0).
Initial exploration confirmed that the dataset is sufficiently large for reliable model training and evaluation.

A] Target Variable Distribution

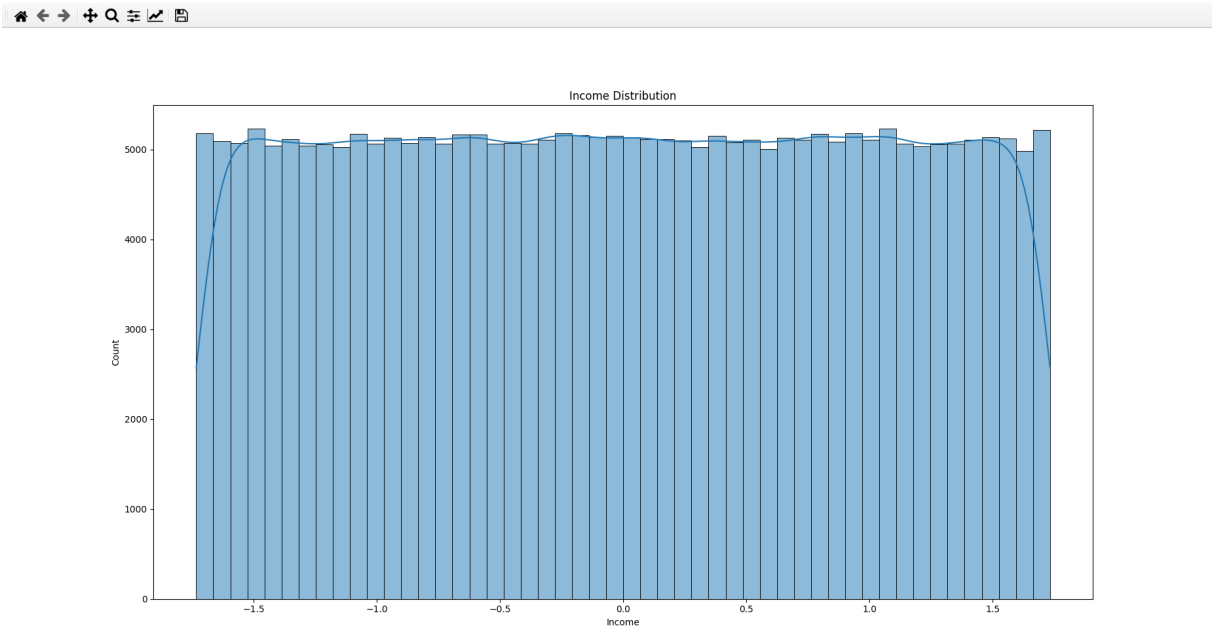
The distribution of the target variable shows that **non-default cases are more frequent than default cases**, indicating some class imbalance in the dataset. However, due to the large dataset size, the model can still learn meaningful patterns. Understanding this imbalance is important because it may influence evaluation metrics such as precision and recall.



B] Distribution of important numerical features

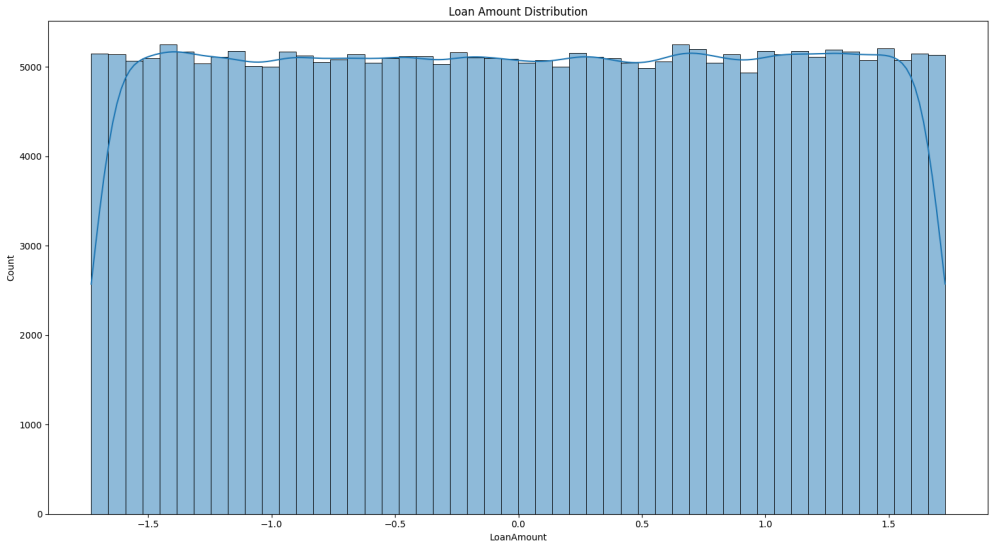
1) - Income Distribution

The distribution of borrower income shows a **wide spread of income levels**, with most applicants concentrated in the lower-to-middle income ranges. This suggests that the dataset represents a diverse set of borrowers across different financial backgrounds.

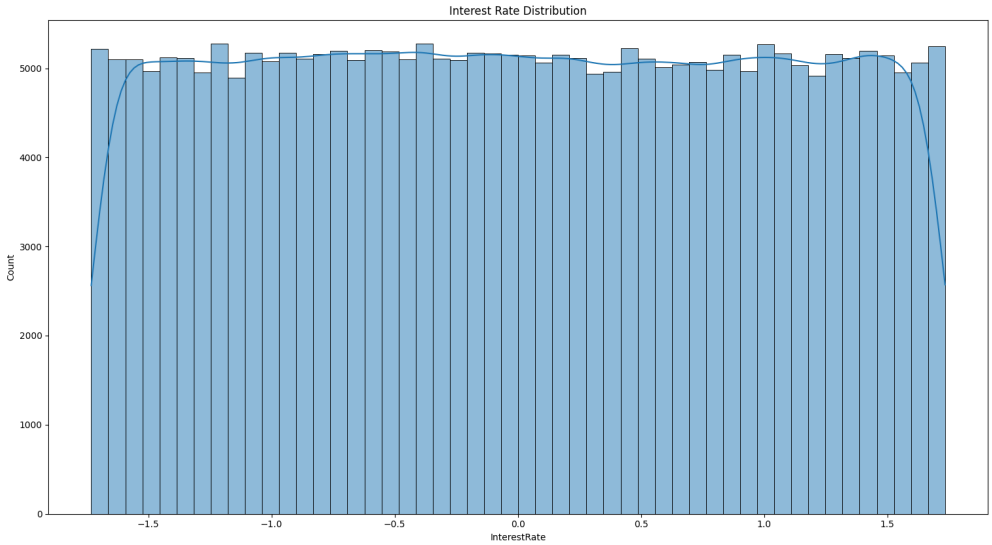


2) - Loan Amount Distribution

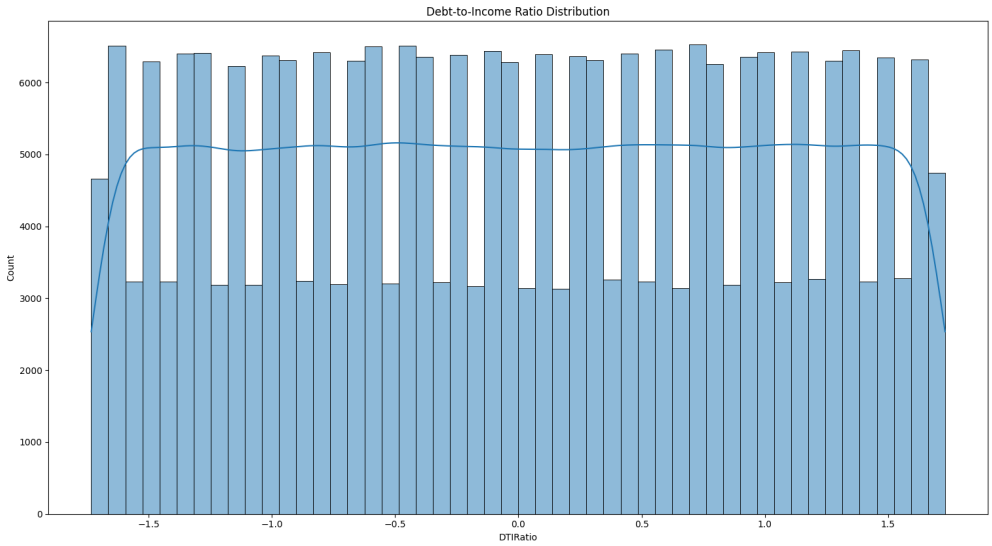
The loan amount distribution indicates that most loans fall within a **moderate borrowing range**, while very large loans are less frequent. This reflects typical lending behavior where borrowers tend to apply for moderate loan amounts relative to their financial capacity.



3) - Interest Rate Distribution

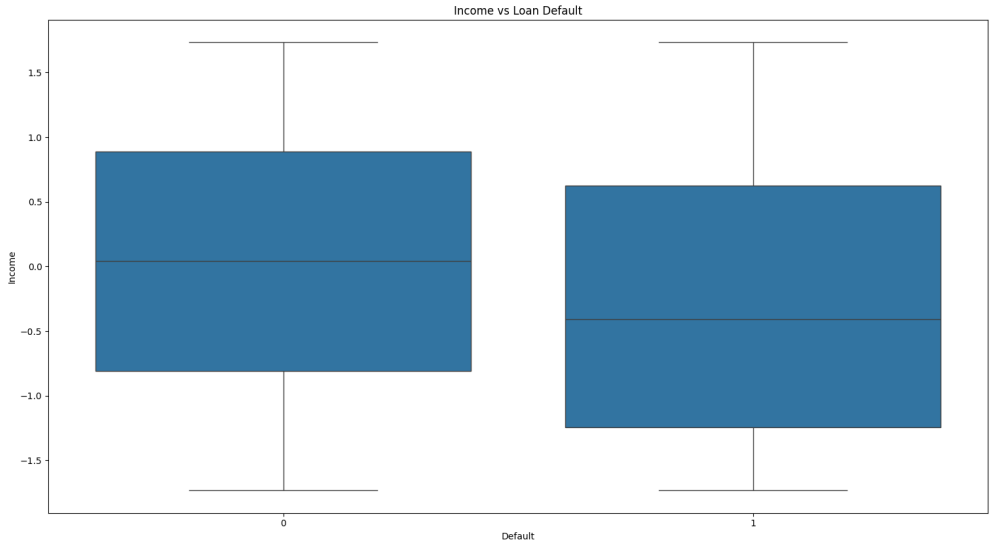


4) - Debt-to-income Ratio Distribiution



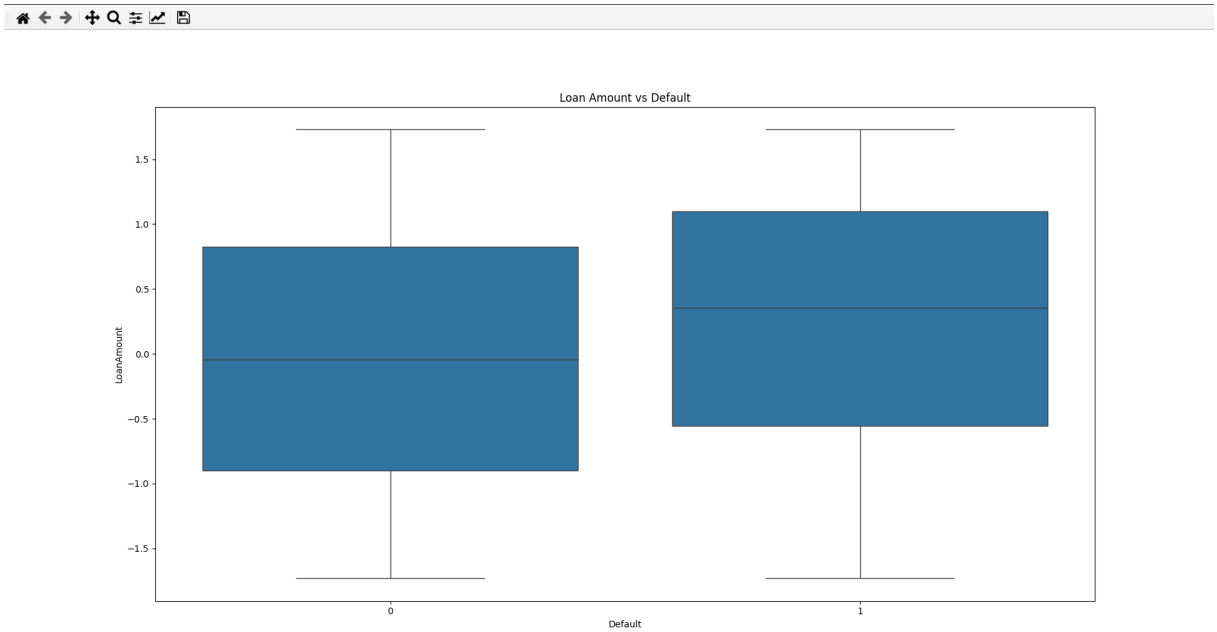
C] Income vs Default

The boxplot comparing income with default status suggests that **borrowers with lower income levels tend to have slightly higher default rates**. This indicates that income level may influence a borrower's ability to repay loans.



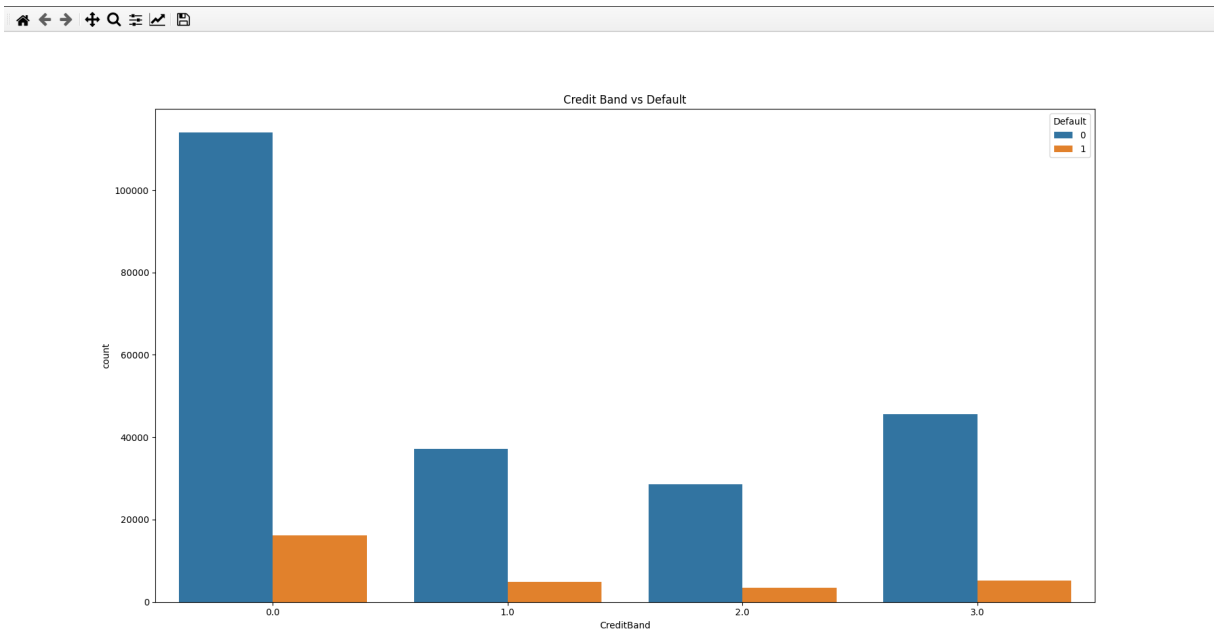
D] Loan Amount vs Default

The comparison between loan amount and default status suggests that **larger loan amounts may slightly increase the likelihood of default**. Higher loan obligations can increase financial pressure on borrowers, making repayment more difficult.



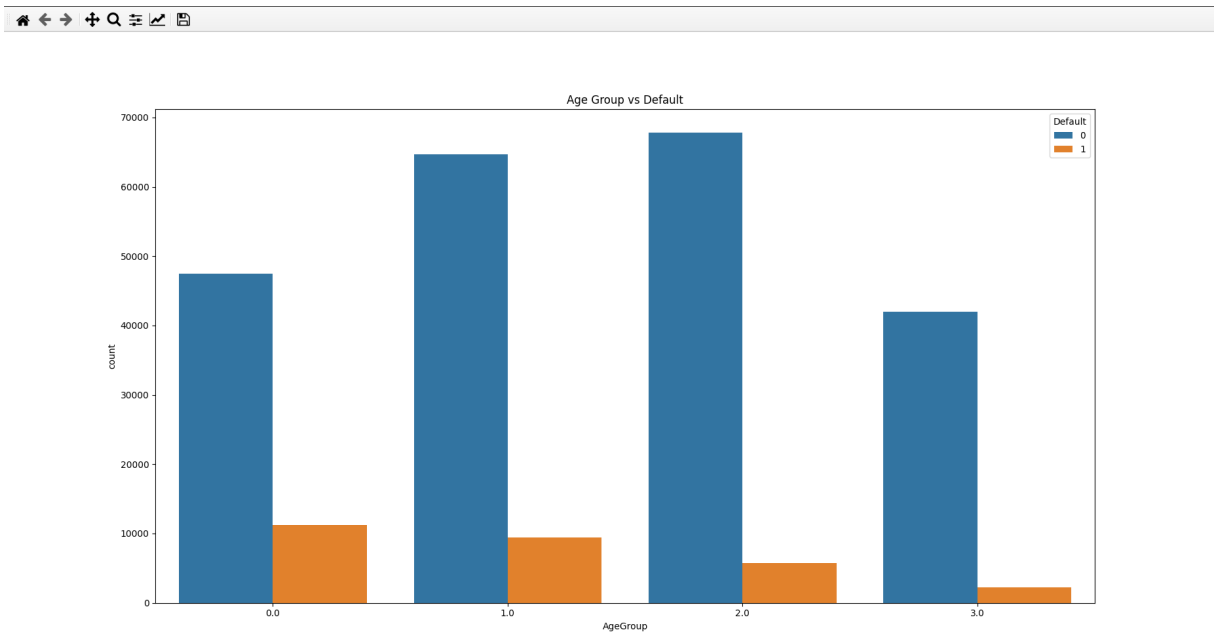
E] Credit Band vs Default

Analysis of credit bands reveals a clear relationship between **credit score category and loan default risk**. Borrowers in lower credit bands exhibit significantly higher default rates compared to those in higher credit bands. This aligns with financial risk theory where credit scores reflect borrower reliability.



F] Age Group vs Default

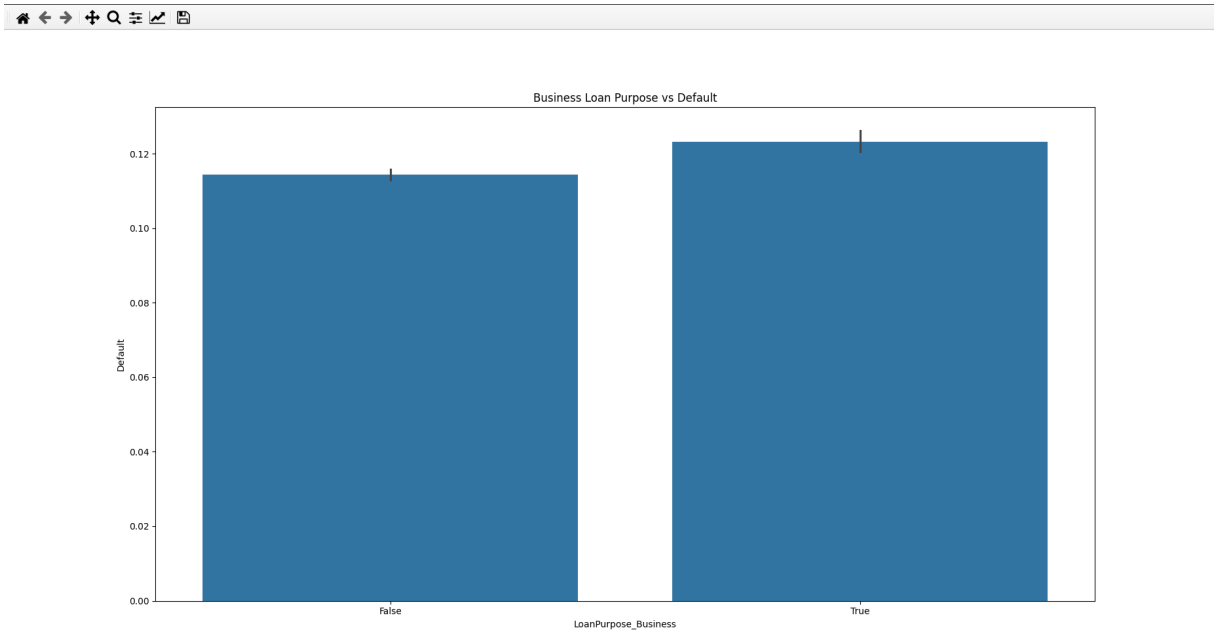
The analysis of default rates across different age groups shows variation in repayment behavior. Younger borrowers may exhibit slightly higher default rates compared to older borrowers, potentially due to lower financial stability or shorter credit histories.



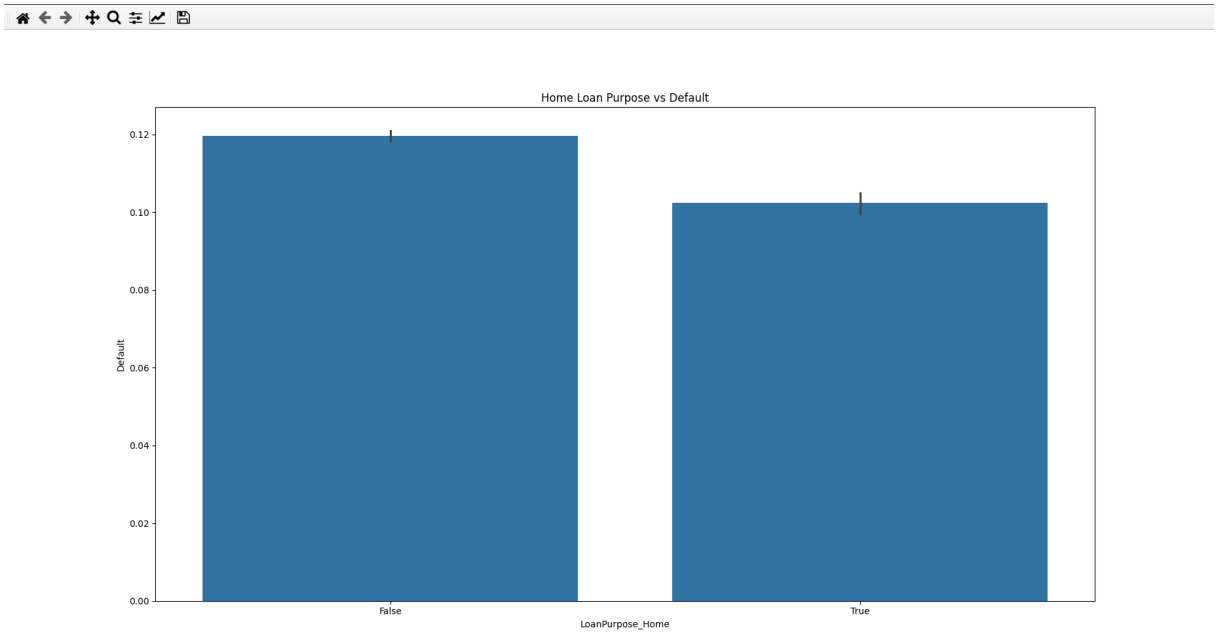
G] Loan Purpose vs Default

Loan purpose also appears to influence default risk. Certain loan purposes may have slightly higher default rates, suggesting that the purpose of borrowing can affect repayment capability.

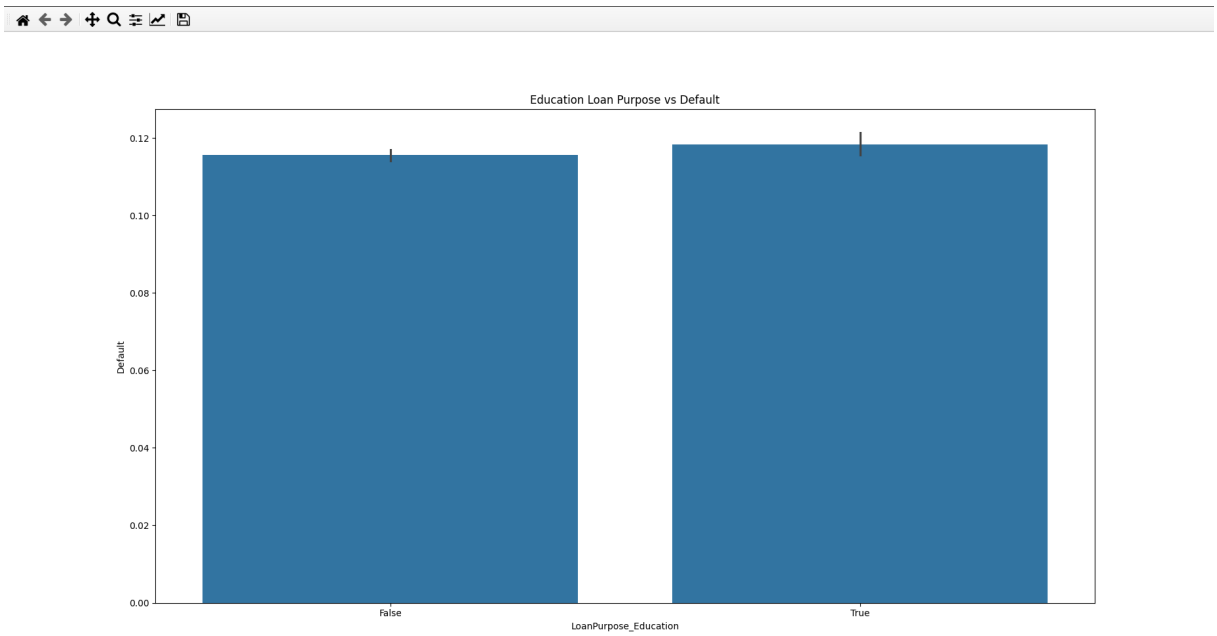
1) Bussiness Loan vs Default :



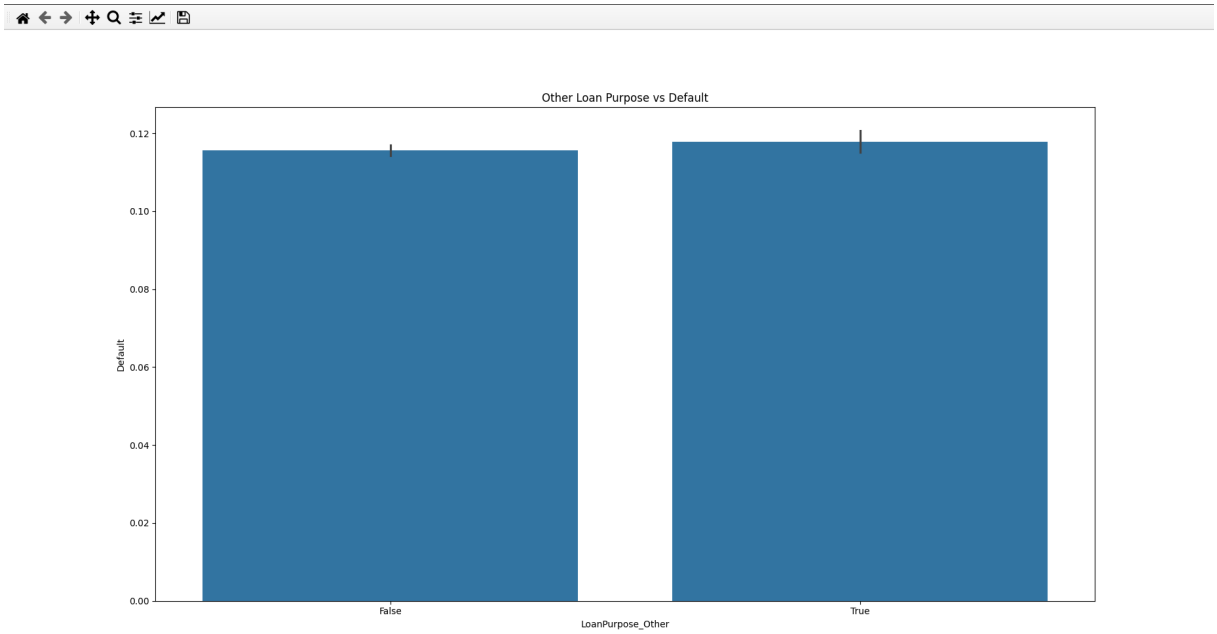
2) Home Loan vs Default :



3) Education Loan vs Default :

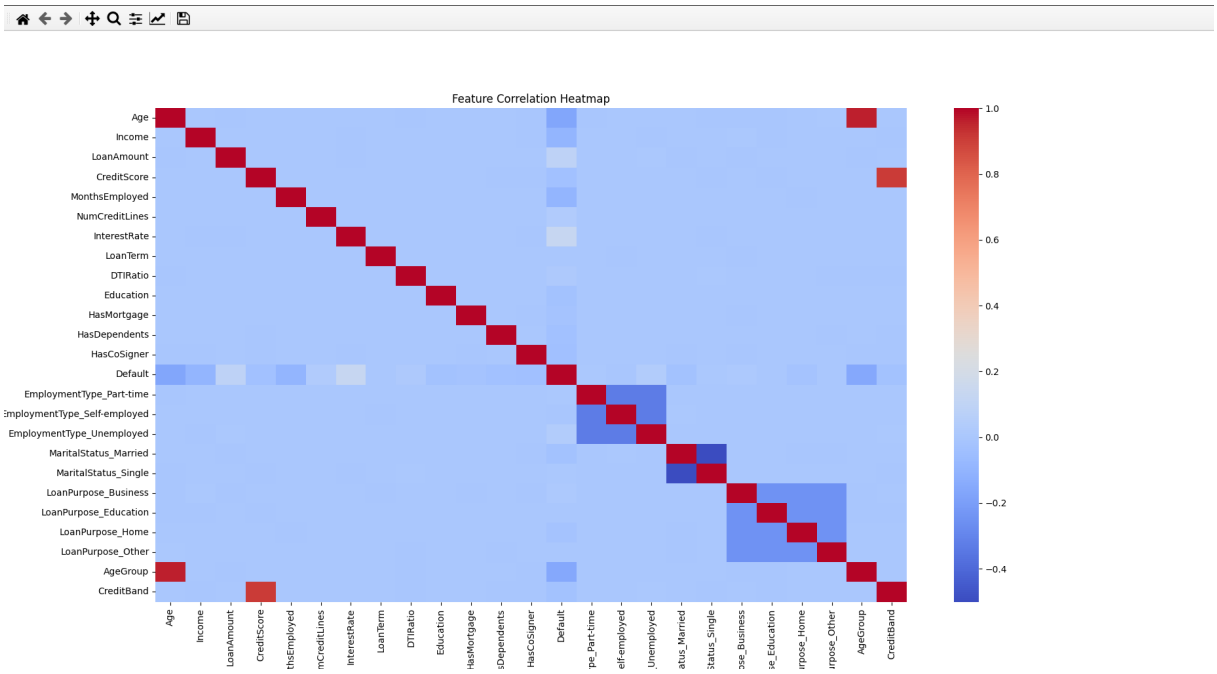


4) Other Loan vs Default :



H] Correlation Analysis

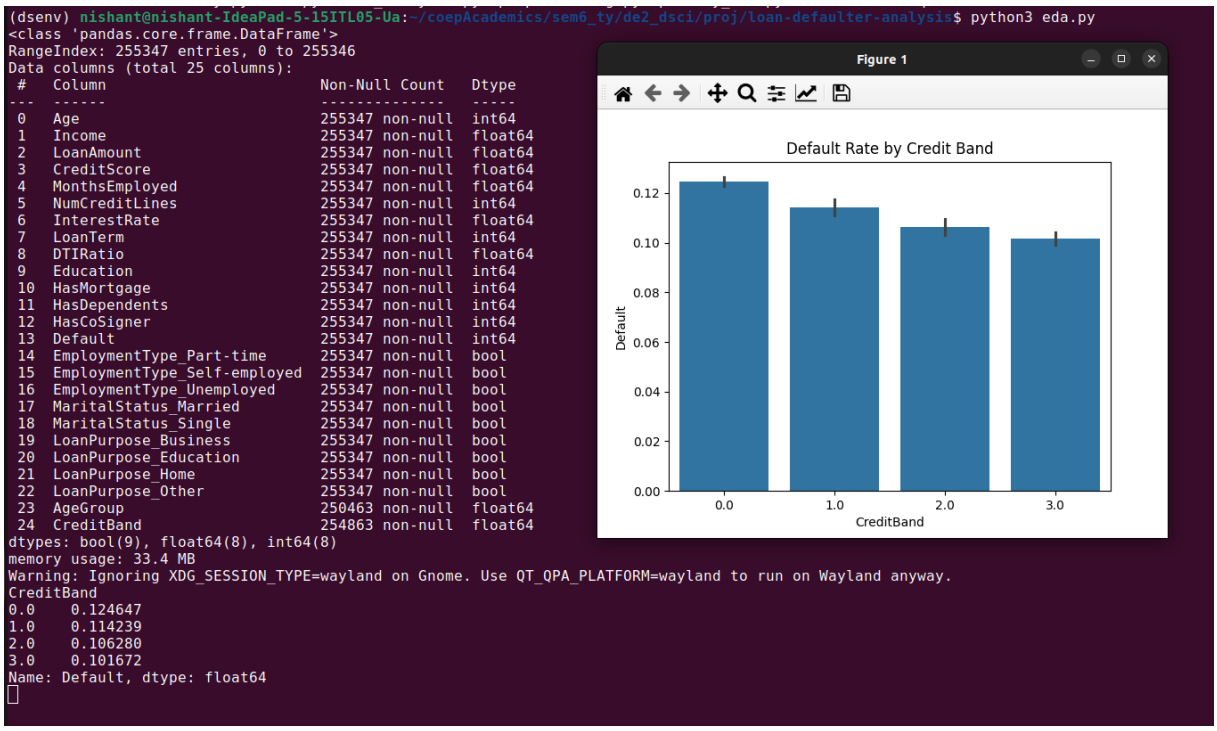
The correlation heatmap highlights relationships between numerical variables. Certain features such as **credit band, debt-to-income ratio, and income** show noticeable relationships with the target variable. These variables are likely to be important predictors for the loan default classification model.



Default Rate by Credit Band — Analysis:

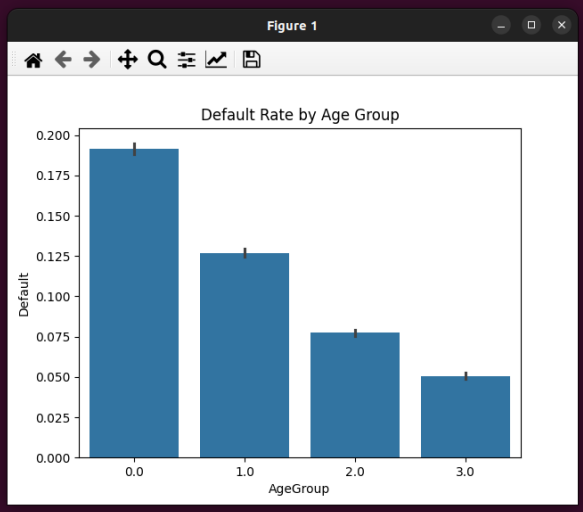
The analysis of default rates across different credit bands shows a clear relationship between **credit score category and loan repayment behavior**. Borrowers belonging to **lower credit bands exhibit significantly higher default rates**, while those in higher credit bands show much lower default probabilities.

This trend is expected because credit scores reflect a borrower's past financial reliability and credit history. Individuals with stronger credit profiles are generally more capable of managing debt obligations. Therefore, **credit band appears to be one of the most important predictors of loan default risk** in the dataset.



Default Rate by Age Group — Analysis:

The analysis of default rates across age groups indicates that **loan repayment behavior varies with borrower age**. Younger borrowers tend to show relatively higher default rates compared to older borrowers. This may be attributed to factors such as lower financial stability, shorter credit history, or less stable employment among younger individuals.

In contrast, older borrowers often demonstrate lower default rates, which may reflect greater financial experience, higher income stability, and stronger credit histories. Thus, **age group appears to have a moderate influence on loan default risk**.

```
(dsenv) nishant@nishant-IdeaPad-5-15ITL05-Ua:~/coepAcademics/sem6_ty/de2_dsci/proj/loan-defaulter-analysis$ python3 eda.py
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 255347 entries, 0 to 255346
Data columns (total 25 columns):
 #   Column                        Non-Null Count   Dtype
---  ------                        --------------   -----
 0   Age                           255347 non-null  int64
 1   Income                        255347 non-null  float64
 2   LoanAmount                    255347 non-null  float64
 3   CreditScore                   255347 non-null  float64
 4   MonthsEmployed                255347 non-null  float64
 5   NumCreditLines                255347 non-null  int64
 6   InterestRate                  255347 non-null  float64
 7   LoanTerm                      255347 non-null  int64
 8   DTIRatio                      255347 non-null  float64
 9   Education                     255347 non-null  int64
 10  HasMortgage                   255347 non-null  int64
 11  HasDependents                 255347 non-null  int64
 12  HasCoSigner                   255347 non-null  int64
 13  Default                       255347 non-null  int64
 14  EmploymentType_Part-time      255347 non-null  bool
 15  EmploymentType_Self-employed  255347 non-null  bool
 16  EmploymentType_Unemployed     255347 non-null  bool
 17  MaritalStatus_Married         255347 non-null  bool
 18  MaritalStatus_Single          255347 non-null  bool
 19  LoanPurpose_Business          255347 non-null  bool
 20  LoanPurpose_Education         255347 non-null  bool
 21  LoanPurpose_Home              255347 non-null  bool
 22  LoanPurpose_Other             255347 non-null  bool
 23  AgeGroup                      250463 non-null  float64
 24  CreditBand                    254863 non-null  float64
dtypes: bool(9), float64(8), int64(8)
memory usage: 33.4 MB
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
CreditBand
0.0    0.124647
1.0    0.114239
2.0    0.106280
3.0    0.101672
Name: Default, dtype: float64
AgeGroup
0.0    0.191522
1.0    0.126953
2.0    0.077428
3.0    0.050541
Name: Default, dtype: float64
```